# Business Intelligence and Reporting

## Management Findings

The MediLink Health Network Business Intelligence and Reporting phase focuses on transforming data from the four healthcare facilities in Kampala, Mukono, Mbarara and Gulu into consolidated management information. The findings below are based on the documented project requirements, database design and identified operational challenges. Since the actual SQL query results have not been provided, these are preliminary management findings, not measured results from the populated data warehouse.

### Finding 1: Fragmented patient information affects consolidated reporting

MediLink previously operated four independent clinic systems, resulting in duplicate patient registrations, inconsistent patient details and difficulty establishing the total number of unique patients served across the network.

The implementation of Master Data Management (MDM), supported by the `m_patient` master table and `patient_source_links` cross-reference table, provides a foundation for consolidating patient identities. The `dim_patient` warehouse dimension enables management to analyse patient demographics without routinely exposing direct identifiers.

### Finding 2: Facility performance requires consolidated monitoring

The four facilities need a common reporting approach to compare outpatient attendance, service utilisation and revenue. The `fact_visits` fact table and `dim_clinic` dimension provide the documented structure for comparing visit volumes and financial measures across facilities.

### Finding 3: Timely monthly reporting is a key operational requirement

The project identifies delays in preparing consolidated Ministry of Health reports as an existing business problem. Manual reporting previously required several staff members approximately two weeks to compile information. MediLink requires consolidated monthly reports covering outpatient attendance, the ten most frequently recorded diagnoses and revenue by service, delivered by the fifth day of each month.

The warehouse and SQL reporting queries provide a basis for producing these reports from consolidated data.

### Finding 4: Data quality must be monitored alongside operational performance

MediLink has identified incomplete and invalid phone numbers, inconsistent patient information and potential duplicate records as data quality concerns. The documented `v_dq_scorecard` view

provides measures of phone completeness, date-of-birth completeness, sex completeness and potential duplicates.

The project sets a phone-number completeness target of at least 97%, with zero invalid phone numbers as a separate quality objective.

### Finding 5: Revenue analysis requires service-level validation

The `fact_visits` warehouse table stores consultation, laboratory and medication amounts in separate columns, alongside total visit revenue. The operational `billing` table records individual billed services, including consultation, laboratory, medication, admission and procedure services.

Consequently, service-level revenue reporting should use the `billing` table when all recorded service categories must be represented. The totals should be reconciled against the warehouse before publication.

### Finding 6: Data privacy must be maintained in management reporting

The project identifies risks involving the disclosure of sensitive patient information. The documented warehouse design uses demographic attributes such as sex, age band and home district to support analysis without requiring direct patient identifiers in routine reports.

## Summary of Management Recommendations

1. Strengthen consolidated reporting: Use the data warehouse as the standard source for network-level performance reports after reconciliation checks pass.
2. Monitor facility performance monthly: Compare attendance, service utilisation and revenue across all four facilities.
3. Enforce data quality targets: Track phone completeness against the 97% target and monitor invalid numbers and potential duplicates.
4. Validate financial reporting: Reconcile service-level revenue from billing records against the warehouse before issuing management reports.
5. Meet the Ministry of Health deadline: Establish a reporting workflow that supports submission by the fifth day of each month.
6. Maintain patient confidentiality: Restrict dashboard access and exclude direct patient identifiers from management visualisations.